

CS211: Computer Organisation
Semester 1, 2025

Assignment #2 (Weight: 10%)

Due: 4:00pm Fiji Time, Friday 9th May, 2025

Learning Outcome: Solve computational problems using assembly and machine language.

Wombat 2

Modify the Wombat 1 computer to Wombat 2 by adding Stack. Stack is a memory which has only one opening where the values are pushed into or popped from stack. The values pushed into the stack will settle down on top of other value. The last value pushed into the stack will be the first one to come out and similarly the first value pushed into the stack will be the last one to come out of the stack. The stack has two operations: which are “Push” and “Pop”. Add a stack, a new stack pointer register, push and pop machine instructions to make Wombat 2 computer.

Also add “call” and “return” machine instructions which will allow subroutine calls. “call” instruction will allow the subprogram to be called (which means start executing the code for the subprogram) whereas “return” instruction will allow the subprogram to end and return to main program (start executing the main program from where function call was made).

Write a complete Wombat 2 assembly language program. The program should solve the problem of assignment 1. There should be three subroutine calls. One to check for input, the second to calculate the $f(n)$ and the third to print the result.

Submission Guidelines

- **This assignment is to be completed individually.**
- Submit two files by zipping which has the wombat program and wombat 2 machine (.cpu) file.
- Only **one** submission is required – rename the file as your id number.
- A **softcopy** of your assignment should be submitted through **Moodle Drop Box**.
- Incorrect/incomplete submissions or any assignments caught as plagiarized work will receive a mark of zero (0).
- Late submissions will **not** be accepted unless prior arrangements have been made with the lecturer and supplemented by documentary evidence.
- Do not submit any piece of assignment work on Moodle Discussion Forums.

The following rubric will be used to mark your assignment. This rubric assumes **no late submission**.

	Unsatisfactory (1)	Satisfactory (2)	Good (3)	Excellent (4)	Enter 1,2,3,4	Weight	Total Score
Machine Design	Machine has stack or stack register	Machine has stack, stack register and machine instruction(s)	Machine has stack, stack register, machine instruction(s) and microinstruction for implementing one machine instruction	Machine has all required components: stack, stack register, machine instructions plus relevant microinstruction		8%	
Program Design	Lengthy program with lots of unused storages, unnecessary instructions. Difficult and inefficient program.	Lengthy program with some unused storages, unnecessary instructions. Program is logical but not an efficient one.	Program has some unnecessary instructions. program is logical and easy to follow	Program has no unnecessary instructions and unused storages. program is efficient, easy to understand		10%	
Comments	some comments included.	Some comments with some description of machine instruction.	Clearly commented with specific purpose of each machine instruction.	Clear and effective comments with description and purpose of each instruction.		2%	

Runtime	Program produces no output	Program produces correct output for some input values.	Program produces correct output for all positive input values.	Program produces correct output for all positive, negative and zero input values.		5%	
---------	----------------------------	--	--	---	--	----	--